Research Collaborator Recommendation

Social Network Analysis AAT

A structural link prediction experiment on Stanford SNAP ca-GrQc

Problem and objectives

Rank currently unconnected researchers using co-authorship structure.

Compare four transparent scores and evaluate missing-link recovery.

Export exact global top 10 and top 20 rankings.

Dataset and graph model

5,242 researcher IDs and 14,484 cleaned co-authorship edges.

Simple, undirected and unweighted graph.

arXiv GR-QC papers: January 1993 to April 2003.

Source: snap.stanford.edu/data/ca-GrQc.html

Four prediction methods

Common Neighbors: number of shared collaborators.

Jaccard: shared / combined collaborators.

Adamic-Adar: sum of 1 / ln(shared neighbor degree).

Preferential Attachment: degree(u) x degree(v).

Reproducible evaluation

Primary seed 42. Random 80/20 split of unique edges.

Keep every node in training, including isolates.

Use one full-graph nonedge per held-out positive.

Score only the training graph. Ties use ascending IDs.

Evaluation metrics

ROC-AUC measures positive/negative ranking quality.

Precision@K = hits / K. Recall@K = hits / test positives.

F1@K is the harmonic mean of precision and recall at K.

Balanced sampled precision differs from all-nonedge precision.

Primary results

Highest primary ROC-AUC: Adamic-Adar (0.88894).

Winner Precision@10: 1.000; Precision@20: 1.000.

Same test results select the winner, without independent validation.

Method	AUC	P@10	P@20
Common Neighbors	0.8887	1.00	1.00
Jaccard	0.8888	1.00	1.00
Adamic-Adar	0.8889	1.00	1.00
Preferential Attachment	0.7350	1.00	1.00

Split sensitivity

Repeat the protocol with seeds 7, 21 and 84.

The primary split remains seed 42.

The exported sensitivity tables report all metrics and mean/SD.

Method	Mean AUC	SD (4 seeds)
Adamic-Adar	0.89772	0.00611
Common Neighbors	0.89747	0.00612
Jaccard	0.89756	0.00607
Preferential Attachment	0.73897	0.00468

Exact collaborator recommendations

All 13,722,177 unordered nonedges are scored.

Four heaps retain only 20 records each, with no candidate filter.

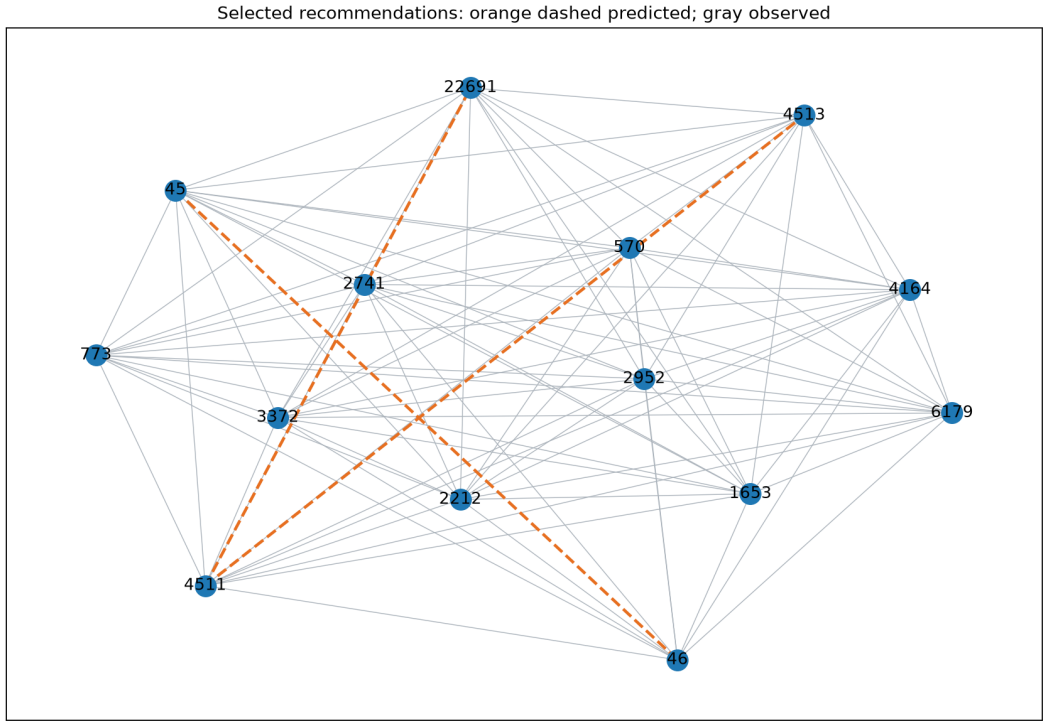
Example (Adamic-Adar): IDs 45 and 46, 42 shared collaborators.

Final recommendations use the full observed graph.

Network visualizations

Bounded subset. Gray = observed; orange dashed = predicted.

Full-network charts, offline HTML and Cytoscape exports accompany this view.



Application demonstration

Open Overview, then select an ID in Network Explorer.

Compare metrics and ROC curves in Algorithm Comparison.

Switch method and top 10/20, explain a pair, download CSV.

Open Researcher Profile for local structure and personal suggestions.

Limitations and conclusion

IDs supply no names, interests or compatibility evidence.

Sampled nonedges are assumed negatives.

Random edge recovery does not prove future prediction.

Structural scores provide explainable possibilities for further study.